



**Verified Carbon
Standard**

UMOYILANGA HYBRID PROJECT

Anthesis 

umoyilanga

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project combines solar, wind and battery storage technologies to offer dispatchable and reliable power to the national electrical grid. The project will operate as a virtual power plant, combining generation from two sites which are over 700 km apart, namely Avondale in the Northern Cape, with 115 MW of solar PV and 30 MW of battery storage, and Dassiesridge in the Eastern Cape, with 63 MW of wind and 45 MW of battery storage.

The unique combination of wind and solar resources with batteries, enables Umoyilanga to provide 75 MW on demand from 05:00 to 21:30 as per the requirements of the PPA, and demonstrates that renewable energy can provide reliable, dispatchable power at a competitive price.

To achieve this, batteries at Dassiesridge will generally charge from the wind energy at night, discharging power in the morning until the sun rises. The solar installation at Avondale will supply the bulk of the energy during the day, supplemented by wind energy from Dassiesridge. Excess solar energy will be used to charge the batteries at Avondale, which will discharge after sunset. A sophisticated energy management system will give instructions to assets across both sites to optimise the power supply in real time, depending on weather forecasts and Eskom's requirements. The low-carbon electricity produced, will help to meet the electricity needs of 120 000 households for 20 years, based on the Eskom residential consumption average of 3 319 kWh/household.

Scenario existing prior to implementation of project activity

The electricity delivered to the grid by the project activity would have been generated by the operation of fossil fuel-based power plants. The project activity helps to avoid CO₂ emission from fossil fuel-based power plants in the absence of the project activity and generate renewable electricity, which displaced part of the electricity otherwise supplied by grid connected power plants.

The estimated annual average CO₂ emission reduction from the project activity is 300,000 tCO₂e/annum. The total CO₂ emission reduction for the first crediting period is 2,100,000 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation</i>	To be completed at verification	VCS	Carbon Check (India) Private Ltd.	To be completed at verification

1.3 Sectoral Scope and Project Type

Sectoral scope¹	01 - Energy industries (Renewable/non-renewable sources)
Project activity type	Project

1.4 Project Eligibility

1.4.1 General eligibility

As per paragraph 2.1.1 of the VCS Standard, version 4.7, the project is eligible under the scope of the VCS Program because this project activity meets three of the listed points, as described in Table 1 below. The project is also eligible under VCS v4.7, table 2.1, as shown in Table 2 below:

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing coal-based energy supplied by the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ² which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

Table 2: Excluded Project Activities under the VCS v 4.7 table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable to this project which is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	This is a grid connected electricity generation activity using a combination of wind and solar photovoltaic (PV) but the exclusion does not apply to the energy storage system (battery storage) which feeds to the grid. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.	Not applicable to this project which

The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO₂e per year.	Not applicable to this project which does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO₂e per year.	Not applicable to this project which does not install or replace electricity transmission lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not reduce HFC-23 emissions.

Pipeline listing and validation deadlines - The project activity was listed before the validation process began. The opening meeting with the VVB will only take place once the project activity has been listed in the project pipeline with a status of 'Under Validation' in the VCS Program project pipeline. Furthermore, validation shall not begin until the 30-day public comment period has begun, and the VVB shall not complete validation until after the 30-day public comment period has ended. As per para 3.8.1 of VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled by this

project activity as the start date of the project activity is anticipated to be 5 May 2025 (being the day that planned generation will be fed into the national grid and national utility sign-off is achieved) and validation will commence before this date, in 2024.

Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Programme for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit, and the Umoyilanga Hybrid project is a single project with multiple locations and project activities and is not part of a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable as project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable as the project is not a transfer project nor a CPA seeking registration.

1.5 Project Design

Project has been designed as:

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This section is not applicable as the Project is not a Grouped project, but a single project with multiple project activity instances at multiple locations. The design of the multiple project activities is described elsewhere in this document.

1.6 Project Proponent

Organization name	Anthesis B.V.
Contact person	Grant Little
Title	Principal Carbon Developer
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Organization name	Umoyilanga (RF) (Pty) Ltd
Contact person	Gregoire de Montgolfier
Title	Project Director
Address	1st Floor Fernwood House, the Oval – 1 Oakdale Road, Newlands, Cape Town, Western Cape, 7800
Telephone	+27 (0)63 638 9763
Email	gregoire.demontgolfier@edf-re.co.za

1.7 Other Entities Involved in the Project

There are no other entities involved in the development of this project.

1.8 Ownership

Umoyilanga (RF) (Pty) Ltd has the legal right to own and operate the Umoyilanga Hybrid facilities, which can be verified through the following documents provided by the project proponent:

- Power Purchase Agreement dated 30 August 2023 which includes all details of the physical locations and property boundaries [1].
- NERSA Generation Licence dated 21 September 2021 [2].

- The Operator has a legal right to use the land with land lease agreements for the duration of the project lifetime, which can be verified through the following documents provided by the project proponent:
 - Avondale Lease Agreement
 - Dassiesridge lease Agreement

Anthesis and Umoyilanga have a legal agreement⁴ in place for the joint ownership of a registered carbon project.

1.9 Project Start Date

Project start date	5 May 2025
Justification	According to Section 3.8 Project Start Date in VCS Standard Version 4.7, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. In this case the projected start date of project activity is the Commercial Operation Date (COD), on which the plant first starts to feed electricity into the national grid. As such, that is the day on which it started generating emission reduction credits. This has been taken as per the Power Purchase Agreement as the date above, when COD is planned.

1.10 Project Crediting Period

Crediting period	☒ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☐ <i>Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)</i>
Start and end date of first or fixed crediting period	05-May-2025 to <u>04-May-2032</u>

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

⁴ Anthesis, Umoyilanga RF Termsheet Commercial Agreement, signed 22 July 2024

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☐ < 300,000 tCO₂e/year (project)
- ☒ ≥ 300,000 tCO₂e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
05-May-2025 to 31-December-2025	200 000
01-January-2026 to 31-December-2026	300 000
01-January-2027 to 31-December-2027	300 000
01-January-2028 to 31-December-2028	300 000
01-January-2029 to 31-December-2029	300 000
01-January-2030 to 31-December-2030	300 000
01-January-2031 to 31-December-2031	300 000
01-January-2032 to 04-May-2032	100 000
Total estimated ERRs during the first or fixed crediting period	2 100 000
Total number of years	7
Average annual ERRs	300 000

1.12 Description of the Project Activity

The project combines solar, wind and battery storage technologies to offer dispatchable and reliable power to the national electrical grid. The project will operate as a virtual power plant, combining generation from two sites which are over 700 km apart, namely Avondale in the Northern Cape, with 115 MW of solar PV and 30 MW of battery storage, and Dassiesridge in the Eastern Cape, with 63 MW of wind and 45 MW of battery storage.

The unique combination of wind and solar resources with batteries enables Umoyilanga to provide 75 MW on demand from 05:00 to 21:30 as per the requirements of the 20-year PPA [1], and demonstrates that renewable energy can provide reliable, dispatchable power at a competitive price.

The PV plant has been designed to minimize visual and noise impacts, as well as to operate safely and assuring a high level of reliability, with low water consumption and only limited maintenance and repair with a lifetime of 25-30 years. No downtime due to maintenance is expected.

PV, Wind and BESS are operated remotely to optimise availability during the dispatchable period, getting most of its energy from solar during the day, and wind during the night. Excess energy is used

to charge the batteries. Over the term of the PPA, Umoyilanga guarantees 84% availability over the dispatchable periods based on solar and wind resources assessment. The plant layout is shown in Figure 1 below.

Umoyilanga - Hybrid Renewable Project

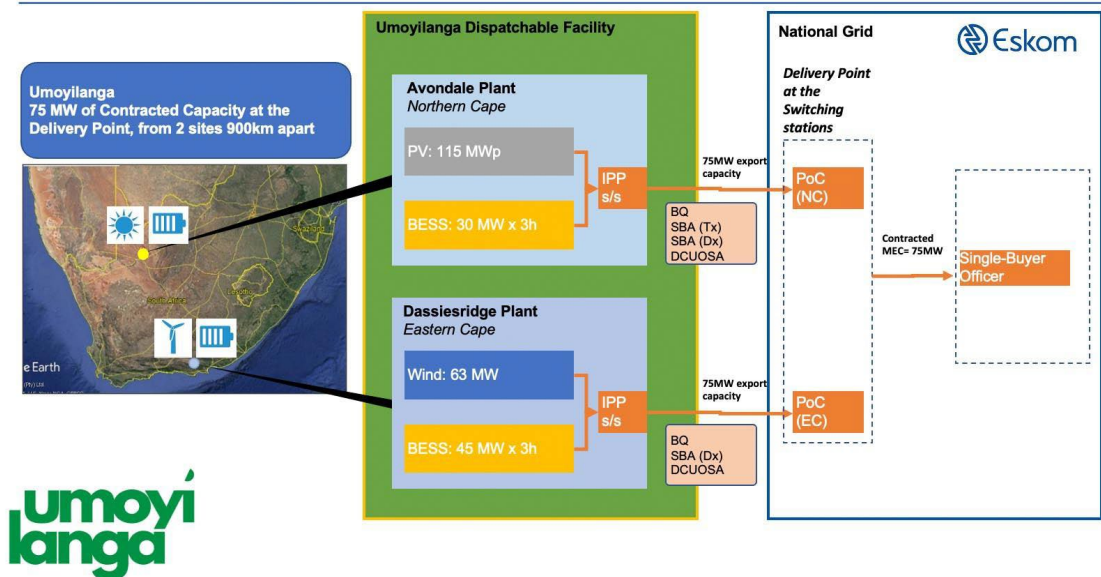


Figure 1: Umoyilanga Plant Layout

The main technical details of Umoyilanga Hybrid Renewable Projects are shown in the tables below.

Table 3: Proposed technologies for PV project activity

Avondale Solar PV Plant (Northern Cape)			
Technology	Type	Manufacturer (proposed)	
Solar PV System			
PV Module	Thin-film PV modules or Mono/polycrystalline PV modules,	Seraphim (35%) Longi (65%)	20-year life time
Inverter	13 x 6,6 MVA 1 x 4,4 MVA	Sungrow	20-year life time
Switchboard	33 kV MV		20-year life time
Transformer	MV/HV 132 kV		
Trackers	1V tracker	Game Change	
BESS			
	90 MWh	Sungrow	20-year life time

Table 4: Proposed technologies for Wind project activity

Dassiesridge Wind Plant (Eastern Cape)			
Technology	Type	Manufacturer (proposed)	
Wind System			
Wind turbines	V150 4,5MW HH120mm	Vestas	20-year life time
BESS	135 MWh	Sungrow	20-year life time
Switchboard	33 kV MV		20-year life time
Transformer	MV/HV 132 kV		20-year life time
Virtual Power Plant Optimiser & Controller (VPPOC)			
VPPOC	SCADA	InAccess (PowerFactors)	

The project implementation schedule is shown in Table 5 below.

Table 5: Project Implementation Schedule

NO.	Activity	Expected Finish Date
1	PV Plant	Feb/25
2	PV Substation and Switchgear	Jan/25
3	PV BESS	Feb/25
4	PV BESS Grid Compliance	Apr/25
5	Wind Plant	Apr/25
6	Wind Substation and Switchgear	Nov/24
7	Wind BESS	Jan/25
8	Wind BESS Grid Compliance	Apr/25
9	VPPOC	May/25

1.13 Project Location

The Umoyilanga project is located at two unique locations within South Africa that are connected via a Virtual Power Plant (VPP). A KML file has been provided in the VERRA registry. The locations are highlighted below.

- The Avondale site lies on Portion 1 of Farm Avondale 410, Gordonia RD within the Dawid Kruiper local municipality, approximately 30 km east of Upington City adjacent to the N14 national highway.
- The Dassiesridge site lies within the Nelson Mandela Bay Municipality (NMBM), approximately 18 km northeast of Uitenhage (Kariega) town, off R75 regional road.

1.14 Conditions Prior to Project Initiation

The project activity is an emission-avoidance project where no emissions will be generated in the construction, operation and maintenance phases of the project. The electricity generation conditions in South Africa, prior to including this CSP plant into the national grid, are described under Section 3.4 (Baseline Scenario).

Although this is not an AFOLU project, the environmental conditions are discussed as this is a Greenfield project and is established on an area previously zoned as agricultural land. The primary purpose of the project is to supply renewable and reliable electricity to South African consumers, and as such, it is not implemented with the aim of generating GHG emissions for the purpose of their subsequent reduction, removal, or destruction. Since the project comprises of a solar plant and a wind plant in two separate locations, the environmental conditions will be discussed for each site of the project. Separate EIAs were conducted for each site and all the associated infrastructure on each site.

Ecosystem Type

The Avondale site, wherein the solar plant will be located, is categorized as being in the Savannah Biome, which is characterized by a combination of a grassy ground layer and scattered woody plants making up the upper layer. Due to the dominant vegetation being grasslands, and the hot and dry weather conditions, the area is prone to natural fires, although infrequent and of low intensity, which maintain the dominant grassland structure of the ecosystem [3].

The Dassiesridge site is located in the Albany Thicket Biome that is a species-rich, evergreen scrubland which is home to a variety of fauna such as springbok, zebra and wildebeest. Notable species include the Eastern Leopard Toad that is endemic to the Eastern Cape, and avifauna such as *Aves-circus maurus* of high sensitivity [4].

Current and Historical Land-use

The current land-use in the Avondale project site is grazing by game and agricultural activities, along with surrounding areas close to the Orange River. The surrounding land uses also include small-scale mining.

The western half of the Dassiesridge project site is currently used for game farming, whilst the eastern half is the de-proclaimed Grassridge Private Nature Reserve, adjacent to the operational Grassridge Wind Energy Facility.

Climate

The climate for the Avondale Solar Park site can be described as warm-temperate, with annual temperatures ranging from -4°C to 41.5°C . The region experiences very dry winters, with the average annual precipitation varying between 120 mm and 260 mm.

The climate for the Dassiesridge site is classified as hot semi-arid with an annual average temperature of 18.2°C and an average annual rainfall of 427 mm.

Topography

The Avondale Solar Park site is located in flat terrain which is suitable for the development of a solar power plant.

The Dassiesridge site topography is characterized by moderate undulating hills, and the site is at an altitude ranging from 180 meters to 300 meters above sea level.

Hydrology

The most prominent hydrological feature of close proximity to the Avondale project site is the Orange River, which runs moderately close to the southern boundary of the project site. The ridge located south of the site is a terrace that defines the floodplain of the Orange River, and drainage occurs as sheet wash towards the low lying area, along the eastern boundary of the study area and through a drainage towards the Orange River.

The Dassiesridge project site traverses a number of non-perennial rivers within the drainage network in the region, hence the site is classified as a Strategic Water Source Area. The western half of the site occurs within 500 m of 5 wetlands, thus triggering the requirement for a Water Use Authorization.

Soils

Based on the soils classification study at the Avondale project site, the soils are described to be weakly structured and sandy, forming a mosaic of Glenrosa soils and very shallow rocky soils. Due to their shallow nature, low pH and climatic conditions, the soils have low agricultural potential, thus making the area unsuitable for crop cultivation under arable conditions.

The soil types at the Dassiesridge site are Eutric Regosols that are typically young soils that are low in organic matter and have a low water-holding capacity, thus making them sensitive to drought and overall unfavourable for agricultural purposes.

Vegetation

Since the Avondale site is located within the Savannah biome, the dominant vegetation type is a lower layer of grassland, along with woody plants spread out across the site. Notable plant species include the *Acacia erioloba* and the *Aloe claviflora* that trigger the requirement of a permit prior to removal of these species during construction.

The Dassiesridge project site traverses two vegetation types, namely the Grassridge Bontveld and the Sundays Valley Thicket which occur exclusively within the Eastern Cape province. These vegetation types both occur in the project site, forming a combination of areas dominated by well-developed woody tree species, and shrubland dominated by Fynbos.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were considered:

Table 6: Relevant Legislation

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), states that everyone has a right to a non-threatening environment and requires that reasonable measures are applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced in NEMA and given further expression.
National Development Plan, Agenda 2030	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Environmental Conservation Act (No. 73 of 1989)	To provide for the effective protection and controlled utilisation of the environment and for matters incidental thereto.

National Environmental Management Act, NEMA. (No. 107 of 1998)	Requires adherence to the principles of Integrated Environmental Management (IEM) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of our water resources.
Conservation of Agricultural Resources (Act 43 of 1983)	Provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water sources and vegetation and the combating of weeds and invader plants; and for matters connected therewith.
Occupational Health and Safety Act (No. 85 of 1993)	The Occupational Health and Safety Act, 1993, requires the employer to bring about and maintain, as far as reasonably practicable, a work environment that is safe and without risk to the health of the workers. This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment, processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The act intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic

	conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labour Organisation; and to provide for matters connected therewith.
Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016	The act establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and peak load new generation capacity; and cross border projects.
Integrated Resource Plan (IRP2019), and the draft IRP 2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the environment (minimize negative emissions and water usage).
Carbon Tax Act 15 of 2019 and supporting regulations	Provide for VCU's to be used to offset carbon tax liability.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 367,000 tons of CO₂ equivalent, totalling 6,300,000 tons of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 385 000 – 400 000 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.

Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. This will be achieved through just less than 890 job opportunities during the construction phase and a number of jobs during the ongoing operation of the project into the future.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7,2,1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 385 000 - 400 000 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 8 000 GWh of emission intensive grid electricity through the generation of low emission renewable electricity.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 367 000 tCO _{2e} per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 6,300,000 tCO _{2e} of emission intensive grid electricity through the generation of low emission renewable electricity over the 21-year project lifetime.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 33.5% over the past decade (2012 to 2022) this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	During the construction phase there will be 890 job opportunities created and during Operation there will be a number of jobs. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services) [5].	The project planned to create 890 job opportunities during the construction phase (from domestic labour) and a further number of permanent jobs during operations [5]. The Operational jobs will be ongoing, but the construction jobs would only be during the construction phase and will phase out over a period once the plant is operational. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiatives [5].

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Use the table below to describe the stakeholder identification process. Where the rows do not apply, provide justification in the cell in the table below.

Stakeholder Identification	An extensive stakeholder engagement process was conducted as part of the environmental impact assessment that took place in the Environmental Impact Assessment (EIA) phase of the project between 2013 and 2020. This included the following aspects of the project that had separate processes: • Avondale Solar PV site • Dassiesridge Wind Energy Facility • Dassiesridge Ancillary infrastructure • Overhead power lines in the Sundays River Valley and Nelson Mandela Bay Municipalities
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The following measures were implemented to identify stakeholders and to request members of the public to register as interested and affected parties (I&APs):

- Notices were placed in various local newspapers (Noordkaap Bulletin for the Northern Cape facility and the UD Express community newspaper for the Eastern Cape facility).
- A background information document was sent to all adjacent landowners.
- A background information document was distributed to stakeholders via email and placed in the local public library in Uitenhage. Die Gemsbok
- Site notices were placed on the boundary of the sites.
- A stakeholder and interested and affected party identification process was undertaken to create a register of parties. A portion of the process was undertaken during the COVID pandemic, so electronic formats such as SMS, email and a dedicated website link.

All interested and affected party (I&AP) details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and responses provided. This database will be updated on an on-going basis. Key stakeholders include representatives from:

- Department of Forestry, Fisheries and the Environment (DFFE)
- Department of Water and Sanitation (DWS)
- South African Heritage Resource Agency (SAHRA)
- Department of Environmental Affairs: Biodiversity & Conservation
- Department of Energy
- Department of Rural Development and Land Reform
- Department of Science & Technology (DST)
- Relevant local and district municipalities
- Parastatals including Eskom and the CSIR

	➤ Square Kilometer Array (SKA) / SA Large Telescope (SALT) ➤ Affected and surrounding landowners ➤ Environmental non-governmental organisations (Endangered Wildlife Trust, Birdlife SA) ➤ Community based organisations ➤ Residents of surrounding communities and towns
Legal or customary tenure/access rights	The legal tenure of the land is private ownership, and a binding lease agreement is in place with the landowners, in order to establish the Umoyilanga Hybrid Plant facilities on the land. Dassiesridge: EDF Renewables, on behalf of Dassiesridge Wind Power (Pty) Ltd, has engaged with the landowner and received written consent to undertake the proposed activities on the proposed property, from the affected landowner.
Stakeholder diversity and changes over time	Key stakeholders include government agencies, neighbouring landowners and residents, local communities, businesses and environmental NGOs. The Dassiesridge site lies within the Nelson Mandela Bay Municipality (NMBM). Social challenges within this municipality include unemployment and poverty. According to the NMBM IDP, job creation and sustainable livelihood form part of the central development needs of this area [4]. The agricultural, conservation, tourism and manufacturing sectors are the key role players in the economies surrounding the project area. As such, these sectors play an important role in the social, cultural and economic fabric of the surrounding communities. The Avondale site lies within the Dawid Kruiper local municipality, where the main economic activities include Agriculture, business services, game farming, tourism and hospitality [6]. Challenges experienced by residents of surrounding communities included work shortage and poverty. Approximately 45 – 50% of the surrounding communities live in poverty [7]. Changes in the make-up of the group are anticipated to be minimal throughout the duration of the project.

Expected changes in well-being	No significant negative impacts on the well-being of stakeholders are anticipated to occur because of this project. However, there are many positive impacts. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiatives. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services) [5]. This will result in: • Increased production / business sales (SMMEs) • Increase in local Gross Domestic Product (GDP) • Increase in employment opportunities • Increase in skills development • Increase in household income • Improved housing and basic service provision.
Location of stakeholders	There are no stakeholders that reside in close proximity to the project. The closest local communities that may be influenced are those of, and these are not in immediate proximity: • Avondale Solar ○ Karos (southern bank of Orange River) ○ Leerkrans (southern bank of Orange River) ○ Upington (limited) • Dassiesridge Wind: ○ Motherwell ○ Kariega/Uitenhage ○ Addo
Location of resources	The Umoyilanga Hybrid facility is located on two facilities. 1. Avondale1 Solar Park - on Portion No. 1 of the Farm Avondale No. 410, Gordonina Registration Division, Dawid Kruiper Local Municipality, ZF Mgcawu District Municipality, Northern Cape Province), 34 km East of Upington, South Africa.

	2. Dassiesridge Wind Energy Farm - is located on the following farm portions: a. Farm 185 (Elands Hoorn), Portion 11; b. Farm 187 (Grassridge); c. Farm 188 (Gringley); d. Farm 189 (Blauw Baatjes Vley), Portion 5; and e. Farm 189 (Blauw Baatjes Vley), Remaining Extent (RE). These properties are situated within the Nelson Mandela Bay Municipality and the Sundays River Valley Local Municipality, within the Eastern Cape Province. The landowners have granted Umoyilanga the rights to lease the land for the generation of electricity for the next 20 years.
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2.1.2 Stakeholder Consultation and Ongoing Communication

The table below describes the process for and the outcomes from the stakeholder consultation conducted prior to project initiation.

<i>Date of stakeholder consultation</i>	11 October 2024
Stakeholder engagement process	A stakeholder engagement process formed part of the scoping and Environmental Impact Assessment (EIA) phase of the project, starting in 2013 and continuing until September 2020. The outcome of this process is a database of interested and affected parties (I&APs) which has been kept up to date and expanded since then [4]. This included stakeholder identification as set out in 2.1.1. Public meetings could not be held at the time due to the COVID 19 pandemic, but appropriate channels and mechanisms were used. The purpose of the consultation was to provide information about the project and to provide stakeholders with a platform to provide input to the project planning process. The communication was presented in a culturally appropriate and inclusive manner, and inclusive language was used.

	The additional stakeholder consultation that was carried out specifically in relation to the VCS requirements, built on the existing stakeholder communication process and activities. All stakeholders in the registered I&AP database were contacted directly via phone calls/ SMS and/or emails, and notices were printed out and put up at the project sites. Advertisements were also placed in local newspapers prior to listing inviting stakeholder and public comment.
Consultation outcome	All issues raised during the public consultation process have been included in the issues and responses report which forms part of the EIA [3], [4] . All issues reported have been resolved and a record of all communication has been kept. In summary, no project or environmental related concerns have been raised by registered I&APs nor have any objections been received in this regard. Additionally, the project has been received very well by all relevant commenting authorities and no objections or opposition has been received. To date, no issues or comments have been received in relation to the VCS validation and verification process.
Ongoing communication	Ongoing communication via direct electronic communication to stakeholders recorded in the registered I&AP database, as well as public notices placed in accessible locations in the surrounding communities. In addition, community liaison officers have been appointed to facilitate communication with local communities on an ongoing basis. All grievances received are recorder, an addressed as far as practically possible.
Stakeholder input	All comments received during the EIA and stakeholder consultation has been carefully considered and addressed in the EIA and project implementation. No updates to the project design have been considered necessary. Stakeholder comments will continue to be monitored to consider whether further updates to the project are required.



Plate 5.1: Proof of Site Notice board placed at 33°35'58.55"S, 25°31'25.84"E.

Figure 2: Example of public notice placed on site

from Kariega to the world for 32 years, and

grounds and sectors.

Quintin Levey (Exporters EC Chairman).

PHOTO: SUPPLIED

Dis heerlike Nartjiefees dié Saterdag!

DIE lekkerste en beste gesinsfees in die Oos-Kaap – die Nartjiefees – vind weer Saterdag, 19 Oktober op Loerie plaas!

Maak seker jy is daar vir al die lekker musiek deur Fatman, Ruan van der Kuil, Dillon Lerm en Chris en Elmarie Brits.

Vir die honger maag is daar smullekker kos, gebak, poeding, pannekoek, tannie Joan se beroemde tuisgemaakte konfyt, plaasvars groente om saam huis toe te neem en natuurlik heerlike nartjies!

Kuier in die saal se koffiewinkel met ou en nuwe vriende terwyl die kleinspan die kindervermaak geniet. Staan nader vir die nartjie-eet kompetisie of snuffel rond by die kunsmarkstalletjies.

Die Nartjiefeesterrein beskik oor veilige parkeerplek, en voorsiening vir gestremde persone is aan die onderkant van die groentestalletjie gemaak. Die hekke open om 08:00 en die fees behoort teen 15:30 klaar te wees.

Toegangsfoeie is R120 vir volwassenes en R80 vir leerders. Voorskoolse kinders – vergesel van hul ouers – kom gratis fees toe. Kaartjies is by die hek en op Quicket beskikbaar.

Kaartfasiliteite is by die fees beskikbaar. Bring opvoustoele saam om na die kunstenaars te luister. Geen plekniekmandjies of gazebo's word op die terrein toegelaat nie.

Verdere inligting is op Facebook en www.nartjiefees.co.za beskikbaar. Ander navrae kan by 083 383 4932 verkry word.

Umoyilanga Project

Umoyilanga Hybrid Renewables Project to boost SA's Clean Power Supply in 2025

The Umoyilanga hybrid renewables battery energy storage project is set to begin operations next year, integrating solar, wind, and battery storage technologies to deliver 75MW of reliable, dispatchable, clean power to South Africa's national electricity grid.

Functioning as a virtual power plant, the project will harness energy from two sites – Avondale in the Northern Cape and Dazzlesridge in the Eastern Cape. It is expected to create approximately 450 full-time jobs during the construction phase.

Owned by EDF Renewables and Perpetua Holdings, Umoyilanga will produce electricity under a 20-year power purchase agreement ratified in August 2023. This initiative is part of the South African government's Risk Mitigation Independent Power Producer Procurement Programme. It will reduce greenhouse gas emissions by replacing fossil fuel derived electricity with clean renewable energy and, as such, deliver carbon credits under the Verified Carbon Standard (VCS). These credits are developed by Anthesis and can be used by companies to offset their carbon tax obligations. Independent auditors will be validating the project to ensure it meets the strict international requirements of the VCS. Independent auditors will be validating the project to ensure it meets the strict international requirements of VERRA.

For more information about this project, or to raise any questions or concerns, please email projectsa@anthesisgroup.com

About Anthesis: Anthesis empowers clients to achieve sustainable performance as the world's leading purpose-driven, digitally enabled, science-based activator. In South Africa, our core offerings include Carbon Project Development, Carbon Tax Advisory, and Carbon Offsetting, all designed to help organizations navigate their environmental responsibilities and drive impactful change.

Loerie Nartjiefees 2024

19 OKTOBER, NG KERK LOERIE

09:00 DILLON LERM,
10:30 CHRIS BRITS
11:30 RUAN VAN DER KUIL,
12:30 NARTJIE-EET
13:15 CHRIS & ELMARIE BRITS,
14:00 FATMAN

HEELTYD FEESTYD, FAMILIE VERMAAK, LEKKER KOS!

Quicket **HEKKE OPEN BAM**
KAARTJIES R120 R80
BESKIBBAAR
YR MEER INLIGTING KONTAK 083 383 4932
WWW.NARTJIEFEES.CO.ZA

Figure 3: Example of Newspaper advert (UD Express - October 2024)

2.1.3 Free Prior and Informed Consent

Obtaining consent	The landowners have consented and granted rights to the project to install the associated energy and storage infrastructure on the properties and lease agreements are in place. No IPs (Indigenous Peoples), LCs (Local Communities) or customary rights holders are impacted by the Umoyilanga Hybrid project activities.
Outcome of FPIC	The outcome of the process is that the Umoyilanga Hybrid facility will be established on private land for the next 20 years. No other land resources or communities have been negatively impacted.

2.1.4 Grievance Redress Procedure

Development process	Grievances can be received via two channels: • The grievance portal on the Anthesis website • There are also permanently employed community liaison officers stationed in the surrounding communities with the goal of keeping communication up to date with Umoyilanga. Any grievances received by Anthesis will be handled on a case-by-case basis by Anthesis in consultation with the Umoyilanga project team. Those by the Umoyilanga team will be communicated with Anthesis periodically.
Grievance redress procedure	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress process consists of: • Receiving the issue from a stakeholder • Assigning it to the relevant manager/team • Responding to the stakeholder (which may include telephonic, written or face to face communication) in order to understand and resolve the issue in good faith • Recording the outcome of the interaction. • Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa. Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged, reviewed and taken into account in ongoing project operation, management reporting and future verifications.

2.1.5 Public Comments

Comments received	Actions taken
This section will be completed once the public commenting period has passed. No comments received at time of publishing.	This section will be completed once the public commenting period has passed

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits for over 2 decades. The Umoyilanga project proponent has extensive renewable energy project experience in South Africa region, including similar Renewable Power facilities at Waainek; Chaba and Wesley-Ciskei. Their parent company is a global power company with experience in projects such as Aramon Solar PV plant, France; Maverick Solar PV plant, USA; Al Dhafra Solar PV plant, UAE; Ventos de Bahai wind, Brazil and Dumat Al Jandal Wind, Kingdom of Saudi Arabia.

The integrated team is well positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No natural and human induced risks	Measures to prevent natural and human-induced risks were taken and documented in the EIA report.
Risks to stakeholder participation	No risks identified	Stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of participation barriers to stakeholders identified. The process for stakeholder's participation was undertaken as

		required by the South African EIA requirements.
Working conditions	No risks identified related to working conditions	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks related to women and girls identified	PP has taken all measures to ensure safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks identified	PP ensures safety of minority and marginalized groups, including children in the project site as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	During Construction phase, increased traffic volumes and people will lead to increased levels of air pollution and noise. During Operational phase, some smoke from burning of waste could cause air pollution.	> Speed of vehicles on roads should be controlled e.g. speed bumps and speed restrictions. > All roads should preferably be sealed to eliminate dust formation caused by strong winds and vehicle movement. > Solid waste should not be burned on the project area. > Fire belts around the development must be made according to the regulations of the Veld and Forest Fire Act. > Vegetation refuse should be composted if possible and re-used.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁵	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Equal pay for equal work	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Child labor	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.
Human trafficking	No risks identified	PP have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.

⁵ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	PP adheres to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Sites of archeological or cultural significance may exist within project boundary	An archaeological-cum-heritage assessment was conducted by AGES to ascertain whether there are any remains of significance in the area that will be affected by the proposed development. The significance of the impact on the heritage resources was determined to be LOW. All the heritage sites found on the property should be excluded from the PV plant footprint and a minimum 30 m buffer zone is recommended.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	All property rights are covered by legally binding lease agreements that cover land tenure and no at-risk communities were involved or affected.

2.3.5 Benefit Sharing

The project does not impact on property rights and therefore no benefit sharing agreement is required

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Potential loss of endangered plant species during construction of pylons and infrastructure.	Botanical specialists will be contracted to identify No-Go Areas and delineate where pylons and infrastructure cannot be installed.
Soil degradation and soil erosion	Clearance of vegetation may increase run-off and lead to top-soil erosion	EMPr includes Effective Monitoring System for erosion risks and revegetation and re-seeding of disturbed areas for erosion control and rehabilitation.
Water consumption and stress	- Pollution of hydrological features due to erosion risks during storm events. - Leakage of sanitation system into groundwater	EMPr includes storm water management infrastructure to be clear from debris to ensure no leakage into groundwater and surrounding hydrological features.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	There is no significant risk to species or habitat that had been identified.
Areas needed for habitat connectivity	The project has not adversely impacted any areas of significance that are needed for habitat connectivity

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	No mitigation /preventive measure taken as there is no threat to Areas for habitat connectivity
Areas for habitat connectivity	No risk identified	No mitigation /preventive measure taken as there is no threat to Areas for habitat connectivity

2.4.2 Introduction of Species

Not applicable, as there is no introduction of new species in the construction or operation of the project activity.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Opuntia ficus-indica</i>	• EMPr to incorporate removal of invasive species through biological control method. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Opuntia aurantiaca</i>	• EMPr to incorporate removal of invasive species through biological control method. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Acacia mearnsii</i>	• EMPr to incorporate removal of invasive species through biological control method. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Acacia cyclops</i>	• EMPr to incorporate removal of invasive species through biological control method. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risks identified	No invasive species are to be introduced. Measures taken are documented in the EMPr.

2.4.3 Ecosystem Conversion

This section is not applicable, seeing that this is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system.	7.0
Tool	TOOL 24	Common practice	3.1
Tool	TOOL 27	Tool for the Investment analysis	14.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s);	The project activity involves installation of a greenfield power plant that is connected to the national grid of South Africa. So, this meets the applicability condition (a).

Methodology ID	Applicability condition	Justification of compliance
	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	This project activity is a grid connected renewable energy power generation project which includes the integration of a BESS into the greenfield power plant. So, the project meets applicability condition (a)
	The methodology is applicable under the following conditions:	

Methodology ID	Applicability condition	Justification of compliance
	(a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions	(a) This project is a hybrid renewable energy plant utilising solar power and wind power. (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase agreement⁶ with the national grid. (d) BESS will only be charged from the greenfield renewable energy (solar and wind) plants at each location. No grid power will be used to charge the BESS component.

⁶ Reference to PPA

Methodology ID	Applicability condition	Justification of compliance
	following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	(e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	(a) Not applicable to this project as the project does not employ any fuel switching (b) Not applicable to this project as the project does not employ any biomass firing
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable to this project as the project does not employ any retrofits, rehabilitations, replacements, or capacity additions

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project activity does not entail proposing a new methodology. As such TOOL 01 is mandatory, and therefore was applied to prove additionality.
CDM TOOL 01	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	TOOL 01 is referenced in the methodology ACM0002 as the prescribed tool to prove additionality and, as such it is mandatory for this project.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This project will feed electricity into the South African national grid, and as such, this is the prescribed tool to calculate the baseline emissions.
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case,	This project does not include any off-grid power plants. Therefore, this tool was used to calculate the emission factor for the project

	two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	baseline electricity system for grid power plants only.
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in a non-Annex I country and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 24	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use	Project activity applies methodological tool: Tool for the demonstration and assessment of additionality and therefore the Common Practice tool is applicable to the project activity.

	the common practice test for the demonstration of additionality.	
CDM TOOL 24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Methodology ACM0002 does not define separate approach for the conduction of common practice test, which are different from the ones on the tool.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 27	This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	Project activity complies as it does use the investment analysis to demonstrate additionality. This is presented in §3.5 below.
CDM TOOL 27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Project activity complies as methodology ACM0002 provides for the use of investment analysis.

3.3 Project Boundary

As per the methodology ACM002 version 22, the project boundary includes the power plants, and all power plants connected physically to the electricity system that the project power plant is connected to.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	Excluded for simplification. Minor source of emissions. Conservative
		Other	No	No other GHG emissions relevant
Project	Greenfield Solar Photovoltaic electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Greenfield Wind electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity from the renewable energy generation plants.	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		CH ₄	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

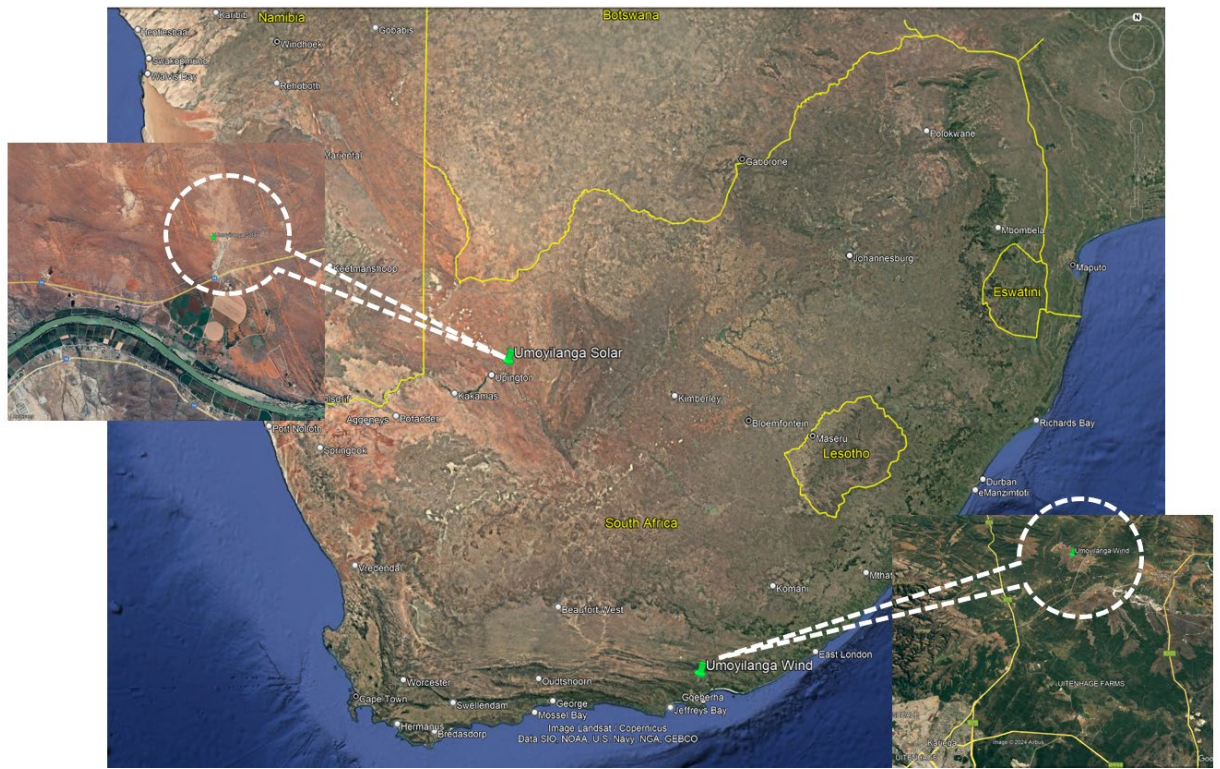


Figure 4: Physical location of the project activities

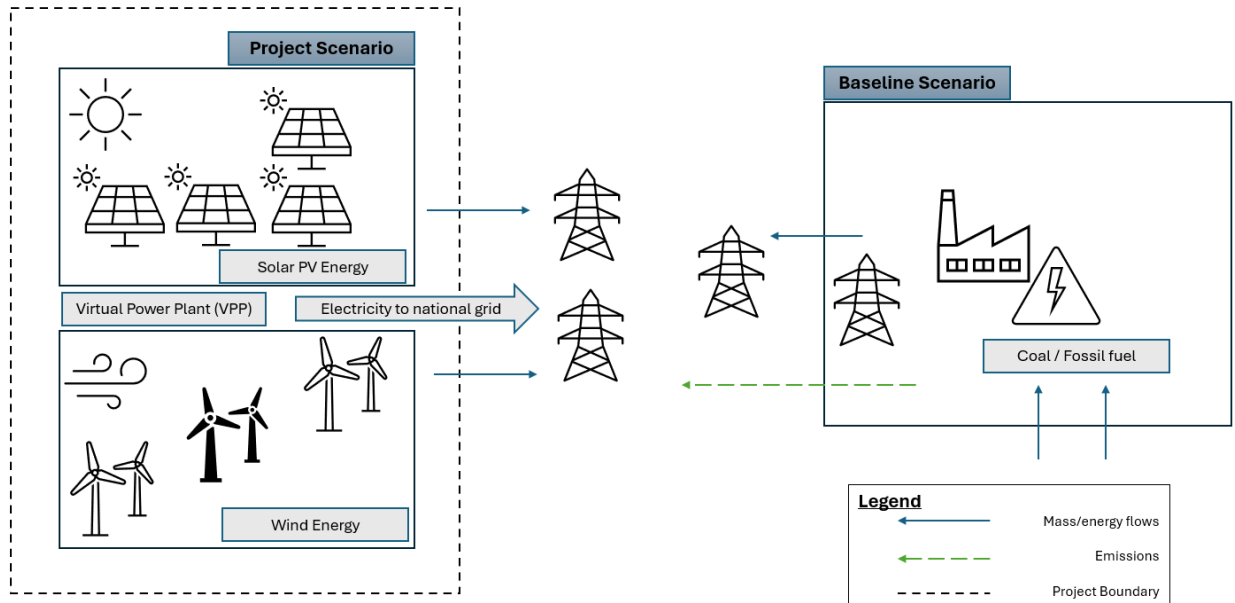


Figure 5 :Project boundary, mass & energy flows of Umoyilanga Hybrid Project

3.4 Baseline Scenario

The baseline scenario for the Umoyilanga Hybrid project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel power plants, predominantly coal, and the addition of new fossil fuel-based generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [11]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [3].

According to South Africa’s Integrated Resource Plan (IRP 2019), coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.959 tCO₂e/MWh over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the Redstone project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL07 “Tool to calculate the emission factor for an electricity system” Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.959 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per AM Tool 07 are outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 7 below:

Table 7: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield Umoyilanga Hybrid generation facilities.
5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind power plant/unit with the integration of BESS	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of two greenfield power plants with associated BESS component. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

3.5 Additionality

Additionality was demonstrated as stipulated in section 5.3 of ACM0002, using CDM tool01 version 7. In addition, Tool 27 version 14 and tool 24 version 3.1 were applied.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the emission reductions achieved by the project would not have occurred in the absence of carbon finance. Tool 27 version 13 prescribes a stepwise approach, as follows:

The input values used in the investment analysis were valid and applicable at the time of the investment decision taken by the project participant. For this project, financial close was reached in February 2024, and the investment analysis was therefore carried out with 2024 values. The steps followed are outlined below:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

Although this is one of two projects being developed in parallel that are the first to combine solar PV, wind and battery storage for electricity generation into the South African power grid, this was not considered in the additionality argument, as there are other solar PV and wind projects of a similar scale within the country.

Step 1: Identification of alternatives to the project activity

Step 1 requires that realistic and credible alternatives to the project activity be defined through the following sub-steps:

Sub-Step 1a: Define alternatives to the project activity

This step entails defining realistic and credible alternatives available to the project participants or similar project developers that provide outputs or services comparable to the proposed project activities. This is shown in the table below:

Project Alternatives	Applicability to this project
Alternative a: The proposed project activity is undertaken without being registered as a VCS project	Realistic scenario: The project proponent could proceed with this project without carbon finance. The project would still produce renewable energy for sale into the South African electricity grid. This is legally and practically an option, but financial risk associated with the project will be increased thereby rendering it financially not a viable alternative.
Alternative b: Other realistic and credible alternative scenarios to the proposed project activity scenario that delivers electricity with comparable quality, properties and application areas, taking into account examples of scenarios identified in the underlying methodology (ACM0002).	Not a realistic scenario: The generation of renewable energy at the project sites from other sources such as hydro, biomass, geothermal or tidal. The project locations are specifically suited to wind and solar respectively and therefore the other alternatives were not considered viable.
Alternative c: Continuation of the current situation (no project activity is undertaken.) This scenario is therefore equal to the baseline scenario where electricity is delivered to the grid through carbon-intensive fossil-fuel derived energy.	Realistic scenario: The grid could continue operating from fossil-fuel powered plants as described in the baseline scenario.

Outcome of sub-step 1a:

Alternative a: The project proponent could not proceed with this project without carbon finance.

Alternative c: Continuation of the current situation (i.e. baseline scenario).

Sub-Step 1a: Consistency with mandatory laws and regulations

The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution. (This sub-step does not consider national and local policies that do not have legally binding status.)

The scenarios identified in sub-step 1a include solar, wind, and fossil-fuel derived electricity currently feeding into the South African grid. All three these technology options are in compliance with all legal and regulatory requirements in South Africa. There are no laws prohibiting the implementation of the scenarios identified, and other projects do exist in the country as mentioned in Step 0 above. In

addition, all applicable environmental laws have been adhered to as indicated by the approved environmental and water use authorisations.

Outcome of sub-step 1b:

The assessment indicates that scenarios listed above are in compliance with mandatory legislation and regulations in South Africa.

The project proponent can now proceed to step 2 or step 3. Project proponents may also select to complete both steps 2 and 3.

Step 2: Investment analysis

This step entails determining whether the proposed project is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs).

The latest version of the “Guidelines on the assessment of investment analysis”, was also considered during this step. Option (b) was selected for analysis which is to conduct a financially attractive option. The following sub-steps were applied:

Sub-step 2a: Determine appropriate analysis method

Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis. The scenarios identified in step 1 generate financial benefits and therefore the simple cost analysis is not suited. The alternative of continuing the current scenario does not involve an investment by the project developer and therefore the investment comparison analysis is not suited. The benchmark analysis (option III) was most suitable and therefore used for this analysis.

Sub-step 2b: Apply Investment benchmark analysis**Identify financial indicator:**

According to Tool 01, version 7, either project IRR or equity IRR can be used. The financial indicator identified as most appropriate for this type of project was identified as pre-tax project IRR. This is a parameter that is used commonly within the market. According to Tool 27 §16, the applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates or Weighted Average Cost of Capital (WACC) are appropriate benchmarks for a project IRR. Local commercial lending rate was chosen as a suitable benchmark, as obtained from the South African Reserve Bank at the time of investment decision. For 2024, a value of 11.75% was used [8].

Sub-step 2c: Calculation and comparison of financial indicators

The project IRR benchmark analysis was carried out based on the financial assumptions made during the project decision making process. Based on the results, the project IRR is assessed to be less than the selected benchmark. This substantiates that the investment is not financially attractive (Project IRR

for the project activity is less than the Benchmark). Therefore, it can be concluded that the project activity is additional & is not the business-as-usual scenario.

The underlying assumptions are tabulated below:

Project Parameter	Value	Unit	Source
Total Capacity	75	MW	PPA [1]
Annual generation	384 617	MWh/yr	Production model (average over 20 years)
Deration in 1st year	0.6%	%	Production model
Deration 2nd year onward	0.7%	%	Production model
Total plant Capex	6 972 760	kZAR	Project Financials
Tariff rate	1690	ZAR/MWh	PPA (weighted average)
O&M Costs	221 638	kZAR/yr	Average over 20 years. Supplier contracts, inflated annually by CPI
Escalation	4.5	%	South Africa CPI ⁷ at time of investment decision
Insurance	19 156	kZAR/yr	Supplier contracts, 1 st year, inflated annually by CPI
Other recurring costs	33542	kZAR/yr	Inflated annually by CPI
Salvage value	27 953	kZAR	Decommissioning report
Commercial lending rate	11.75	%	SA Reserve Bank

Based on the assumptions above, the pre-tax project IRR would be 6.8% which is below the benchmark of 11.75%

Sub-step 2d: Sensitivity Analysis

A sensitivity analysis was conducted to demonstrate that the project IRR value remains below the benchmark threshold, even when there are variations to the critical input assumptions, as required by the Tool. A range of +10% and -10% was applied. The following variables are considered in the sensitivity analysis:

- Capital cost
- Operations and maintenance costs
- Electricity tariff
- Annual generation

⁷ CPI = Consumer Price Index

Parameter	Project IRR at -10%	Project IRR at +- 0%	Project IRR at + 10%
Capex	8.01%	6.8%	5.75%
O&M	7.14%	6.8%	6.45%
Tariff	5.12%	6.8%	8.35%
Annual generation	5.12%	6.8%	8.35%

The results of the sensitivity analysis indicate that none of the input variances put the project at risk of exceeding the benchmark project IRR.

In conclusion, the proposed project activity requires additional revenues from the sale of carbon credits to be economically attractive to project developers and surpass or get closer to the benchmark.

Step 3: Barrier analysis

§28 of Tool 01 allows for project proponents to proceed to either Step 2 (Investment analysis) or Step 3 (Barrier analysis). Project proponents may also select to complete both Steps 2 and 3. Accordingly, Step 2 is selected. Step 3 Barrier analysis was not deemed necessary and was not considered in this analysis in accordance with the requirements of the Tool.

Step 4: Common practice analysis

The common practice analysis was carried out as stipulated in Tool 24 version 3.1. This tool outlines a step wise approach, which is described below:

Step 1: Calculate applicable capacity output range as +/- 50% of the total design capacity.

The design capacity for the Umoyilanga project is 75 MW and therefore the capacity range for this analysis ranges from 37.5 MW to 112.5 MW.

Step 2: Identify similar projects (both VCS and non-VCS) which fulfill all of the following conditions:

Criteria	Applicability to this project	Projects identified
The projects are located in the applicable geographical area;	The applicable area is the geographical borders of South Africa.	All renewable energy projects listed were considered for this analysis.
The projects apply the same measure as the proposed project activity;	The measure applied is renewable energy fed into the South African electricity grid.	5729,7 MW of renewable energy is supplied to the South African grid (extracted from Eskom's website) [9].
The projects use the same energy source/fuel and feedstock as the proposed	The fuel source is a combination of solar and wind with integrated battery storage.	There is only one similar project meeting this criteria in South

project activity, if a technology switch measure is implemented by the proposed project activity;		Africa and it is currently under development [10].
The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;	Renewable electricity	Solar PV and Wind Energy with BESS
The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;	37.5 MW to 112.5 MW	The capacity of the other project is 128 MW and therefore it falls outside the scope N = 0
The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.		The other project is not yet commissioned and therefore not valid.

It can therefore be concluded that $N_{ALL} = 0$.

Steps 4 and 5 of this section are not applicable since $N_{ALL} = 0$

It can therefore be concluded that the proposed project activity is not common practice within South Africa.

3.6 Methodology Deviations

There are no methodology deviations applied in this project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as set out in AM tool 07, and the calculation steps are described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EGPJ,y \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EGPJ,y$ = Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

The results obtained are: 268,658 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	268,658	t CO ₂ / yr
$EGPJ,y$	280,145	MWh/yr
$EF_{grid,CM,y}$	0,959	t CO ₂ /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 500 000 MWh / yr (from technical specifications)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with Tool07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

4.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd. This is shown schematically in Figure 12 below.

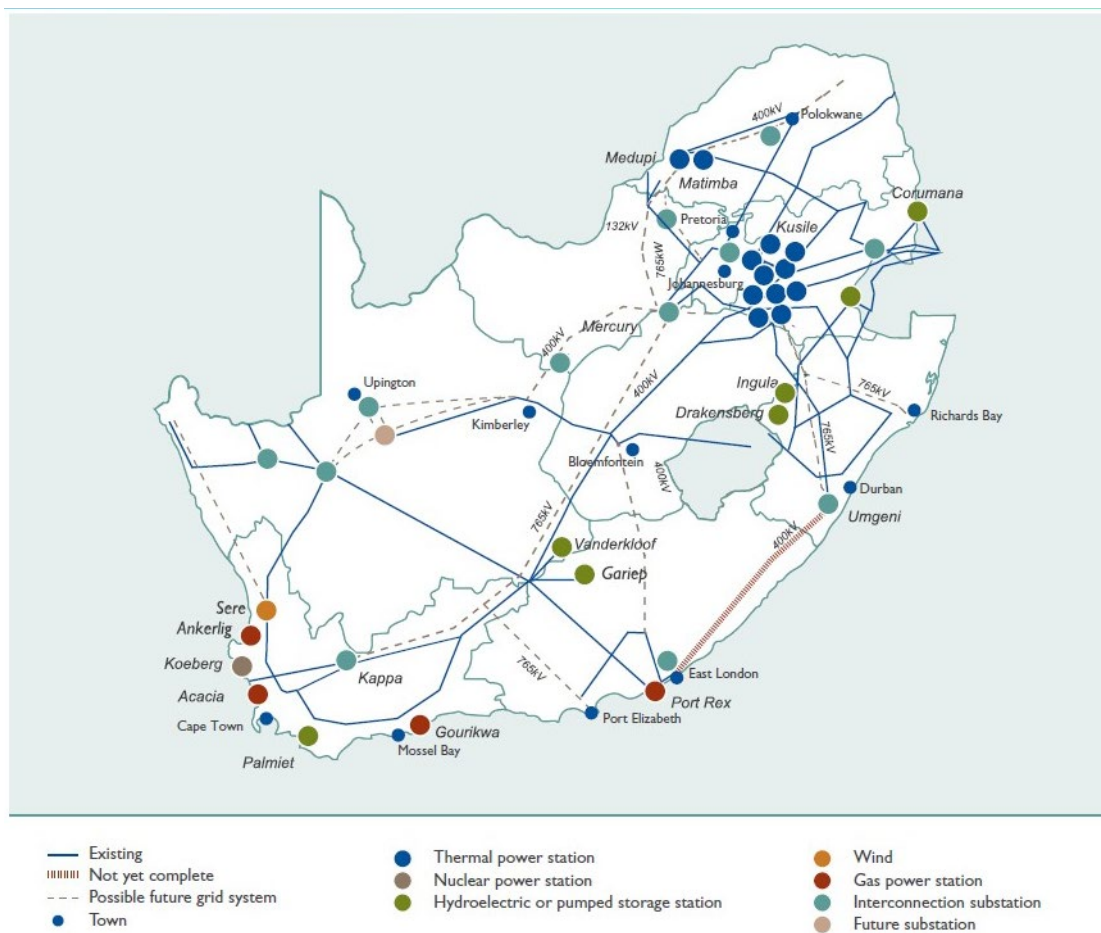


Figure 6: Project electricity system

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country, therefore the grid emission factor is not zero.

4.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 10). were included in this calculation.

4.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (§38.a in TOOL07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL07 40(a) Approach 1, by Equation 3 below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \dots, \frac{EG_{LCMR(y)}}{total_y} \right]$$

Equation 3 (Tool07, 40 Equation 1)

Where:

$Share_{LCMR}$ = Share of the low cost/must run resources (%)

EG_{LCMRy} = Electricity generation supplied by LCMR plants in year y (MWh)

$total_y$ = Total electricity supplied to the project electricity system in year y (MWh)

Y = The most recent year for which data is available

The ex-ante option (TOOL07 §42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2023. Therefore, the generation weighted average of 2021, 2022, and 2023 was used.

The results obtained are shown in Table 8 below

Table 8: LCMR Analysis

Parameter name	Value	Unit
$Share_{LCMR: 2023}$	23,1%	Per cent
$Share_{LCMR: 2022}$	21,3%	Per cent
$Share_{LCMR: 2021}$	19,4%	Per cent
$EG_{LCMRy: 2023}$	49 691	GWh / yr
$EG_{LCMRy: 2022}$	48 092	GWh / yr
$EG_{LCMRy: 2021}$	42 495	GWh / yr
$total_y: 2023$	215 318	GWh / yr
$total_y: 2022$	226 226	GWh / yr
$total_y: 2021$	219 423	GWh / yr

4.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 9 LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
- Gariep - Vanderkloof - Mbashe - First Falls - Ncora - Second Falls	Hydroelectric power stations	Renewable (low generation costs)
- Drakensberg - Ingula - Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
- Acacia - Ankerlig - Gourikwa - Port Rex	Gas and liquid turbine peaking stations	Operated only during hours of highest daily, weekly or seasonal loads (must run)
Sere	Wind energy stations	Renewable (low generation costs)
Koeberg	Nuclear power stations	Nuclear listed as LCMR
- Biomass - Concentrated solar power - Gas/liquid fuel - Hydroelectric - Landfill - Solar PV - Wind	Imports and REIPPP plants	Imports or renewable

The power plants used in the OM calculation are shown in Table 10 below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL07 paragraph 47, option A), as per Equation 4.

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (Tool 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.008 \text{ tCO}_2/\text{MWh}$

All values used are shown in Table 10.

Table 10: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
Arnot	10 148 766	9 085 064	7 530 158	1,17	1,16	1,14
Duvha	11 230 400	9 822 489	9 555 266	1,11	1,13	1,13
Hendrina	4 873 506	4 738 179	2 136 620	1,35	1,24	1,20
Kendal	15 106 544	15 139 746	15 750 630	1,02	1,05	1,04
Kriel	11 328 488	12 597 519	10 727 877	1,32	1,28	1,22
Lethabo	21 832 979	22 890 481	21 367 191	1,00	0,97	0,95
Matimba	23 533 126	23 180 336	23 257 630	1,02	1,04	0,99
Majuba	21 857 835	19 823 301	15 540 628	1,19	1,13	1,09
Matla	18 560 297	18 412 664	15 170 449	1,30	1,27	0,99
Tutuka	10 554 111	8 524 715	7 564 337	1,18	1,23	1,22
Camden	2 518 438	6 670 341	7 378 745	1,10	1,16	1,25
Grootvlei	2 992 859	2 445 162	2 416 293	1,21	1,27	1,24
Komati	708 862	608 728	369 330	1,01	1,03	1,06
Kusile	8 553 383	7 992 189	9 119 476	1,01	1,30	1,09
Medupi	19 753 895	22 637 675	23 246 212	0,88	0,92	0,91
Total	183 553 489	184 568 589	171 130 842			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (Tool 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 10 above.

The values used in Equation 5 are shown in Table 11 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports.

Table 11: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7	21,43	40,40	0,09	0,08
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1	20,39	40,40	0,09	0,08
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8	20,51	40,40	0,09	0,08
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4	17,87	40,40	0,09	0,08
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9	21,64	40,40	0,09	0,08
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7	14,53	40,40	0,09	0,08
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6	19,15	40,40	0,09	0,08
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1	19,58	40,40	0,09	0,08
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0	18,19	40,40	0,09	0,08
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3	19,83	40,40	0,09	0,08
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3	20,45	40,40	0,09	0,08
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3	20,01	40,40	0,09	0,08
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0	19,72	40,40	0,09	0,08
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0	19,42	40,40	0,09	0,08
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9	19,42	40,40	0,09	0,08

4.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL07 §72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2023 was 184 165,63 GWh.

The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) was identified as shown in Table 12. The annual generation of $SET_{5 \text{ units}}$ was 35 365,31 GWh in 2023, and this set comprises 19,2% of AEG_{total} .

Table 12: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2023
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Grootvlei	2008/04/30	GWh/yr	2416,29
4	Komati	2009/03/31	GWh/yr	369,33
5	Sere Wind	2015/03/01	GWh/yr	214
TOTAL				35 365,31

The set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} was determined next, and $SET_{\geq 20\%}$ generated 38 379,02 MWh as shown in Table 13 below, comprised 20,8% of AEG_{total} .

Table 13: Set of power plants that contribute 20% to the Eskom fleet power generation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Ankerlig	2007/03/01	GWh/yr	1332,49
4	Gourikwa	2007/07/01	GWh/yr	1681,22
5	Grootvlei	2008/04/30	GWh/yr	2416,29
6	Komati	2009/03/31	GWh/yr	369,33
7	Sere Wind	2015/03/01	GWh/yr	214,00
TOTAL				38 379,02

SET>20% comprised the larger annual electricity generation and therefore forms the sample set SET_{sample}, but it did contain plants that were commissioned more than 10 years ago and therefore step (d) is followed. Units that were commissioned more than 10 years ago were excluded from the set (those shaded grey in Table 13), and CDM, Gold Standard (GS), and VCS projects were included until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system. This resulted in a set that comprised 20% of the project electricity system and therefore steps (e) and (f) were not required. The resulting SET_{sample-CDM} group of power units, m, used to calculate the build margin is shown

Table 14: Plant set for BM Calculation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
Eskom Fleet Plants				
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Sere Wind	2015/03/01	GWh/yr	214,00
CDM, VCS and Gold Standard Plants				
4	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	GWh/yr	375,17
5	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	GWh/yr	94,53
6	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	GWh/yr	225,86
7	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	GWh/yr	130,30
8	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	GWh/yr	80,30
9	Longyuan Mulilo 1	2012/01/01	GWh/yr	308,14
10	Longyuan Mulilo 2 de Aar, 144 MW	2012/01/01	GWh/yr	497,38
11	Bokpoort CSP (Concentrating Solar Power) Project South Africa	2016/01/01	GWh/yr	200,00
12	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	GWh/yr	878,44
13	Caledon Wind Farm in South Africa	2012/06/01	GWh/yr	406,01
14	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	GWh/yr	140,45
15	Indwe Wind Project	2012/06/01	GWh/yr	148,09
16	Grid Connected Wind Power Plant in Witberg	2012/02/01	GWh/yr	507,20
17	Red Cap Kouga Wind Farm	2011/11/02	GWh/yr	290,00
TOTAL				4281,85

As the annual electricity generation of this set comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group ($SET_{sample-CDM}$) is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO_2/MWh) for power units m during 2023, which is the most recent year y for which electricity generation data is available, using Equation 6:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (Tool 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$ (MWh)	=	Net quantity of electricity delivered to the grid by power unit m in year y
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.81$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 10, $EG_{m,y}$ was used as shown in Table 14.

4.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (Tool 07 85 Equation 16)

As per TOOLI07, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.75$
- $w_{BM} = 0.25$

The results obtained are: $EF_{grid,CM,y} = 0,959$

4.2 Project Emissions

Project emissions are calculated as follows, in accordance with ACM0002 version

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Equation 8 (ACM0002 Equation 1)

Where:

PE_y	=	Project emissions in year y (tCO ₂)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (tCO ₂)
$PE_{GP,y}$ the	=	Project emissions from the operation of geothermal power plants due to release of non-condensable gases in year y (tCO ₂ e/yr)
$PE_{HP,y}$ e/yr)	=	Project emissions from water reservoirs of hydro power plants (tCO ₂ e/yr)

The project does not involve geothermal power plants or water reservoir of hydro power plants therefore the equation simplifies to:

$$PE_y = PE_{FF,y}$$

This plant will not use fossil fuels for electricity generation, only for emergency backup, therefore $PE_{FF,y}$ can be neglected, in accordance with ACM0002 paragraph 42. Thus project emissions are zero.

4.3 Leakage Emissions

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected, in accordance with ACM0002.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The project's emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Equation 9 (ACM0002 Equation 17)

Where:

ER_y = Emission reductions in year y (t CO₂/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
05-May-2025 to 31-December-2025	200 000	0	0	200 000	0	200 000
01-January-2026 to 31-December-2026	300 000	0	0	300 000	0	300 000
01-January-2027 to 31-December-2027	300 000	0	0	300 000	0	300 000
01-January-2028 to 31-December-2028	300 000	0	0	300 000	0	300 000
01-January-2029 to 31-	300 000	0	0	300 000	0	300 000

December-2029						
01-January-2030 to 31-December-2030	300 000	0	0	300 000	0	300 000
01-January-2031 to 31-December-2031	300 000	0	0	300 000	0	300 000
01-January-2032 to 04-May-2032	100 000	0	0	100 000	0	100 000
Total	2 100 000	0	0	2 100 000	0	2 100 000

5 MONITORING

5.1 Data and Parameters Available at Validation

The table below provides all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period.

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data Sources: Records published on Eskom’s website monthly for each power station to fulfil the Atmospheric emission license (EAL) requirements. Eskom’s integrated annual report for 2023
Value applied	1.008
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (tCO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom’s coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL07 “Tool to calculate the emission factor for an electricity system”, based on the net electricity generation and a CO ₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y

Source of data	Calculated using Equation 6, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data source:
Value applied	0.811
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL07 “Tool to calculate the emission factor for an electricity system”. An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data Source:
Value applied	0,959
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL07 “Tool to calculate the emission factor for an electricity system”. Where WOM is equal to: 0.75 and WBM is equal to: 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share _{LCMR}								
Data unit	Percent (%)								
Description	Share of the low cost/must run resources								
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [11] and records published on Eskom’s website [12] for each power station.								
Value applied	Annual LCMR (%) <table><tr><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>19,4%</td><td>21,3%</td><td>23,1%</td></tr></table>			2021	2022	2023	19,4%	21,3%	23,1%
2021	2022	2023							
19,4%	21,3%	23,1%							
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8								
Purpose of data	To calculate baseline emissions, using Equation 3.								
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.								

Data / Parameter	EG _{LCMR,y}						
Data unit	MWh						
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y						
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and records published on Eskom's website [14] for each power station.						
Value applied	<table border="1"> <tr> <td>EG_{LCMRy}: 2023</td><td>49 691 000</td></tr> <tr> <td>EG_{LCMRy}: 2022</td><td>48 092 000</td></tr> <tr> <td>EG_{LCMRy}: 2021</td><td>42 495 000</td></tr> </table>	EG _{LCMRy} : 2023	49 691 000	EG _{LCMRy} : 2022	48 092 000	EG _{LCMRy} : 2021	42 495 000
EG _{LCMRy} : 2023	49 691 000						
EG _{LCMRy} : 2022	48 092 000						
EG _{LCMRy} : 2021	42 495 000						
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8						

Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	$total_y$		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly records published on Eskom's website [xx] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and records published on Eskom's website [12] for each power station		
Value applied	$total_y$: 2023	215 318	GWh / yr
	$total_y$: 2022	226 226	GWh / yr
	$total_y$: 2021	219 423	GWh / yr
	$total_y$: 2023	215 318 000	
	$total_y$: 2022	226 226 000	
	$total_y$: 2021	219 423 000	
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8As presented in Table 8		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.		

Data / Parameter	Y
Data unit	Year
Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [xx] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and records published on Eskom's website [12] for each power station
Value applied	2023
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO ₂ e/MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and records published on Eskom's website [12] for each power station
Value applied	1,.008
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The simple OM emission factor is calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants. using Equation 3 under Option A from section 48 in Tool 07.

Data / Parameter	EG _{m,y}			
Data unit	MWh			
Description	Net electricity delivered to the grid by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [11] and monthly records published on Eskom’s website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [11] and records published on Eskom’s website [12] for each power station			
Value applied	As per Table 10			
	Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023
	Arnot	10 148 766	9 085 064	7 530 158
	Duvha	11 230 400	9 822 489	9 555 266
	Hendrina	4 873 506	4 738 179	2 136 620
	Kendal	15 106 544	15 139 746	15 750 630
	Kriel	11 328 488	12 597 519	10 727 877
	Lethabo	21 832 979	22 890 481	21 367 191
	Matimba	23 533 126	23 180 336	23 257 630
	Majuba	21 857 835	19 823 301	15 540 628
	Matla	18 560 297	18 412 664	15 170 449
	Tutuka	10 554 111	8 524 715	7 564 337

	Camden	2 518 438	6 670 341	7 378 745
	Grootvlei	2 992 859	2 445 162	2 416 293
	Komati	708 862	608 728	369 330
	Kusile	8 553 383	7 992 189	9 119 476
	Medupi	19 753 895	22 637 675	23 246 212
	Total	183 553 489	184 568 589	171 130 842
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10			
Purpose of data	To calculate baseline emissions, using Equation 4.			
Comments	The generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in Tool 07.			
Data / Parameter	$EF_{EL,m,y}$			
Data unit	tCO ₂ /MWh			
Description	CO ₂ emission factor of power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.			
Value applied	As per Table 10			

	Plant Name	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
	Arnot	1,17	1,16	1,14
	Duvha	1,11	1,13	1,13
	Hendrina	1,35	1,24	1,20
	Kendal	1,02	1,05	1,04
	Kriel	1,32	1,28	1,22
	Lethabo	1,00	0,97	0,95
	Matimba	1,02	1,04	0,99
	Majuba	1,19	1,13	1,09
	Matla	1,30	1,27	0,99
	Tutuka	1,18	1,23	1,22
	Camden	1,10	1,16	1,25
	Grootvlei	1,21	1,27	1,24
	Komati	1,01	1,03	1,06
	Kusile	1,01	1,30	1,09
	Medupi	0,88	0,92	0,91
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 10).			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.			

Data / Parameter	$FC_{i,m,y}$
Data unit	Tonnes
Description	Amount of fuel type i consumed by power unit m in year y

Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements

Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements

Value applied

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9

As per Table 11

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 11
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	NCV _{i,y}		
Data unit	GJ/tonne		
Description	Net calorific value of fuel type i in year y		
Source of data	DFFE for coal [15] IPCC default factors for heavy fuel oil , [16]		
Value applied	For coal as per plant in Table 11		
	Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)
	Arnot	21,43	40,40
	Duvha	20,39	40,40
	Hendrina	20,51	40,40
	Kendal	17,87	40,40
	Kriel	21,64	40,40
	Lethabo	14,53	40,40
	Matimba	19,15	40,40
	Majuba	19,58	40,40
	Matla	18,19	40,40
	Tutuka	19,83	40,40
	Camden	20,45	40,40
	Grootvlei	20,01	40,40
	Komati	19,72	40,40
	Kusile	19,42	40,40
Medupi	19,42	40,40	

Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	184 165,63
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10.
Purpose of data	To calculate baseline emissions
Comments	

Data / Parameter	$SET_{5\ units}$
Data unit	Dimensionless
Description	The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil

	the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	35 365,31
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 19,2% of AEG_{total} in 2023.

Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, excluding power units registered as CDM project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [4] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set \geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)

Value applied	38 379,02
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20,8% of AEG_{total} in 2023.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5 \text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, and exclude power units that started to supply electricity to the grid more than 10 years ago
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 32,579,69 GWh in 2023, comprising 17,7% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set \text{ sample}}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding

	those commissioned more than 10 years ago, and that comprise 20% of AE_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	32,579,69
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14 and Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of AE_{total} in 2023.

Data / Parameter	$SET_{sample-CDM}$
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, excludes power units that started producing electricity more than 10 years ago, and including power units registered as CDM project activities, that comprise 20% of AE_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Shown in Table 14.
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14

Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 36 861,54 GWh in 2023, comprising 20,0% of AEG_{total}

Data / Parameter	$AEG_{SET-sample-CDM}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly and annual records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	36 861,54
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated $\geq 20,0\%$ of AEG_{total} in 2023.
Data / Parameter	w_{OM}
Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	Tool07 paragraph 86 (a)
Value applied	75%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7

Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 75% for the first crediting period and for subsequent crediting periods.
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Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	Tool07 paragraph 86 (a)
Value applied	25%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 25% for the first crediting period and for subsequent crediting periods.

Data / Parameter	BE_y
Data unit	tCO ₂ e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1
Value applied	To be determined ex-post: 500 000
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.
Purpose of data	To calculate baseline emissions, using Equation 1
Comments	No additional comments

5.2 Data and Parameters Monitored

The table below provides all data and parameters that will be monitored during the project crediting period. The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

Data / Parameter	$EG_{facility,y}$																		
Data unit	MWh/Year																		
Description	Quantity of net electricity generation supplied by the project (Solar) plant to the grid in year y																		
Source of data	Electricity reading meters																		
Description of measurement methods and procedures to be applied	Electricity generated is metered automatically, recorded electronically on the DCS and SCADA systems, recorded monthly, following the Plant's Standard Operating Procedures (SOPs), and checked against the Eskom meter (located at the independent Noko switching station).																		
Frequency of monitoring/recording	Continuous monitoring and monthly reporting.																		
Value applied	Ex ante values for the first crediting period: <table border="1"> <tr> <td>05-May-2025 to 31-December-2025</td><td>200 000</td></tr> <tr> <td>01-January-2026 to 31-December-2026</td><td>300 000</td></tr> <tr> <td>01-January-2027 to 31-December-2027</td><td>300 000</td></tr> <tr> <td>01-January-2028 to 31-December-2028</td><td>300 000</td></tr> <tr> <td>01-January-2029 to 31-December-2029</td><td>300 000</td></tr> <tr> <td>01-January-2030 to 31-December-2030</td><td>300 000</td></tr> <tr> <td>01-January-2031 to 31-December-2031</td><td>300 000</td></tr> <tr> <td>01-January-2032 to 04-May-2032</td><td>100 000</td></tr> <tr> <td>Total</td><td>2 100 000</td></tr> </table>	05-May-2025 to 31-December-2025	200 000	01-January-2026 to 31-December-2026	300 000	01-January-2027 to 31-December-2027	300 000	01-January-2028 to 31-December-2028	300 000	01-January-2029 to 31-December-2029	300 000	01-January-2030 to 31-December-2030	300 000	01-January-2031 to 31-December-2031	300 000	01-January-2032 to 04-May-2032	100 000	Total	2 100 000
05-May-2025 to 31-December-2025	200 000																		
01-January-2026 to 31-December-2026	300 000																		
01-January-2027 to 31-December-2027	300 000																		
01-January-2028 to 31-December-2028	300 000																		
01-January-2029 to 31-December-2029	300 000																		
01-January-2030 to 31-December-2030	300 000																		
01-January-2031 to 31-December-2031	300 000																		
01-January-2032 to 04-May-2032	100 000																		
Total	2 100 000																		
Monitoring equipment	Electricity meter, (to be confirmed during installation and commissioning), Inaccess Distributed Control System (DCS), which will operate and run the Virtual Power Plant (VPP).																		
QA/QC procedures to be applied	Internal Operational QA/QC for meter reading via the SCADA system have been developed and will meet globally acceptable industry norms. Specifics of the QA/QC will be codified during the																		

	implementation and commission of the plant and included at verification.
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Project emissions in year y
Source of data	Quantified using Equation 8
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoir of hydro power plants. Therefore, project emissions for this renewable energy power generation project will be zero.

Data / Parameter	ER_y
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Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	To be calculated ex-post – Ex-ante value is 500 000 per annum
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emission reductions from the proposed project activity during the crediting period, using available ex ante values. These will be revised during each verification using the actual data recorded (where applicable).

5.3 Monitoring Plan

Understanding that the project is still in design and implementation phase and has not been constructed and final equipment and instrumentation is still to be installed and implemented. This is the best understanding at the time of validation and will be amended and corrected in an “ as-built” version of the plant.

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

Electricity generated by the facility will be monitored internally using an InAccess Distributed Control System (DCS), as part of an overall Supervisory Control and Data Acquisition (SCADA) system which will be used for operational control and reporting as part of the VPP. Apart from reporting for this project,

the data will also be used for billing the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the SCADA system design.

Measurement data will be managed by the project's DCS. The measuring equipment will be installed in three steps:

- Calibration under standard temperature and pressure conditions
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data will be stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment will strictly comply with their specified maintenance plans and calibration certificates will be kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing substation, adjacent to the generating turbines. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon. A check meter, owned by the national utility (Eskom) located at the Eskom Switching Stations which serves as the grid tie-in point. Meters at both locations will have a back-up meter installed as a failsafe.

The meters on site, will be calibrated in accordance with the manufacturer's specifications, and will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records will be kept on site. Calibrations will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474 [17]. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a dedicated data and instrumentation team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organisations will conduct calibrations.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent switching stations. Any discrepancies will immediately

trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the switching stations which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both project personnel and Eskom personnel prior to billing and payment.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information is presented in this document

APPENDIX 2: BIBLIOGRAPHY

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